

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-29 13:06 UTC

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Generated 2026-04-29 13:06 UTC

- Cycles recorded: **229**
- Time span: 135.4h (~1.69 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	206
FALSIFIED	14
BARRIER_HIT	5
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier analysis of boolean functions	3
Noncommutative geometry	2
Representation theory of symmetric groups	2
Ergodic Circuit Framework (COMM_COMPLEXITY)	2
Matroid Theory	2
Algebraic Topology	2
Tropical geometry	2
Algebraic Geometry	2
Random Matrix Theory	2
Additive Combinatorics	2
matroid theory	2
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	2
Combinatorial homotopy theory	1
Kazhdan-Lusztig theory of Hecke algebras	1
Combinatorial algebraic topology	1

Field	Cycles
Knot theory (quantum invariants)	1
Quadratic forms over finite fields (Grothendieck-Witt groups)	1
Clifford algebras over finite fields	1
Toric geometry (via Gröbner degenerations)	1
Arithmetic geometry (Tate-Shafarevich groups)	1
Geometric Langlands correspondence	1
Algebraic graph theory (chromatic polynomials)	1
Clifford algebras over real vector spaces	1
Algebraic cycles (Chow groups over finite fields)	1
Algebraic lattice theory (Möbius functions of flow lattices)	1
Directed algebraic topology (d-space homology)	1
Motivic integration (Denef-Loeser zeta functions)	1
Matroid theory (Clifford index of binary matroids)	1

Last 15 cycles (chronological)

Time	Verdict	Title
2026-04-29 02:17 UTC	INCONCLUSIVE	SOS Hierarchy Integrality Gap for Random CSPs
2026-04-29 02:49 UTC	INCONCLUSIVE	Hardness of MCSP Under Parameterized Circuit Classes
2026-04-29 03:21 UTC	INCONCLUSIVE	Monotone CLIQUE Lower Bound via GF(2) Rank Defect
2026-04-29 03:54 UTC	INCONCLUSIVE	Dual Code Distance Lower Bound for Tseitin Refutations
2026-04-29 04:23 UTC	INCONCLUSIVE	Width-5 ABPs and NC ¹ Circuit Simulation
2026-04-29 05:07 UTC	INCONCLUSIVE	AC ⁰ PARITY Lower Bound via Linear Independence Invariant
2026-04-29 05:27 UTC	INCONCLUSIVE	Representation-Theoretic Rank Invariant for Permanent vs Determinant
2026-04-29 05:58 UTC	INCONCLUSIVE	Submodular Matroid Rank Defect for Monotone CLIQUE
2026-04-29 06:32 UTC	INCONCLUSIVE	SOS Integrality Gap and Secant Variety Dimension for Polynomial C
2026-04-29 07:23 UTC	INCONCLUSIVE	Protocol-Induced Covers on Tensor-Powered Girth Gadgets
2026-04-29 07:53 UTC	INCONCLUSIVE	Exhibit L
2026-04-29 08:50 UTC	INCONCLUSIVE	Real Rank Lower Bound for ACC ⁰ Circuit Size
2026-04-29 10:34 UTC	INCONCLUSIVE	Free Entropy Lower Bound for Disjointness Communication Matrices
		Roe-Cover Multiplicity Lower Bound from Asymptotic Dimension Grow

Time	Verdict	Title
2026-04-29 10:45 UTC	INCONCLUSIVE	Additive Energy of Sipser Function Bounds ACC ⁰ Circuit Size
2026-04-29 13:06 UTC	INCONCLUSIVE	Protocol-Induced Covers on High-Asdim Gadget Lifts Require Expone

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```